

Candidate's Name _____

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MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2011
Higher 2

H2 BIOLOGY

9648/02

Paper 2 Core Paper

13 September 2011**2 hours**Additional Materials: Answer papers

ANSWER SCHEME

Section A

Answer **all** the questions in this section.

QUESTION 1

Fig. 1.1 below depicts an eukaryotic cell.



Fig. 1.1

(a) Describe the function of the region labelled 'X'. [1]

- Transcription of ribosomal RNA gene
- Ribosome assembly
[R: ribosome synthesis, as the protein component is produced in the cytoplasm]

(b) Describe two differences between the products of transcription from regions X and Z. [2]

Area	Region X (nucleolus)	Region Z (euchromatin)
Product of transcription	<u>Ribosomal RNA / rRNA</u>	<u>Messenger RNA / mRNA</u>
Fate of product	Combine with <u>proteins</u> to form <u>ribosomes</u> that are <u>transported to the cytoplasm/RER</u>	Undergoes <u>modification</u> , exit the nucleus <u>into the cytoplasm</u>
Function of product	Serve as a <u>component of ribosomes</u> for translation	Serves as a <u>template</u> for translation

(c) Outline three ways that protein synthesis can be controlled at region Y. [3]

[Ways to promote or prevent translation]

- Initiation of translation can be blocked by regulatory proteins that bind to specific sequences or structures within the 5' end, thus preventing attachment of ribosomes.
- Phosphorylation/presence/activation of translational initiation factors required to initiate translation
[A: reverse argument]
- Post-translational modification occurs to yield functional proteins.
[A: reverse argument]

- Cytoplasmic enzymes can extend poly-A tail prevent degradation of mRNA / prolong translation / initiate translation in embryo
- Polyribosomes at region Y/ rough endoplasmic reticulum can translate an mRNA at the same time to increase the rate of translation / protein synthesis
[R: capping, as that occurs in the nucleus.]
- Availability of amino acids in the cytosol
- Shortening of poly-A tail, resulting in the removal of 5'cap, leading to degradation of mRNA.

Protein synthesis in a prokaryote is illustrated in Fig. 1.2.

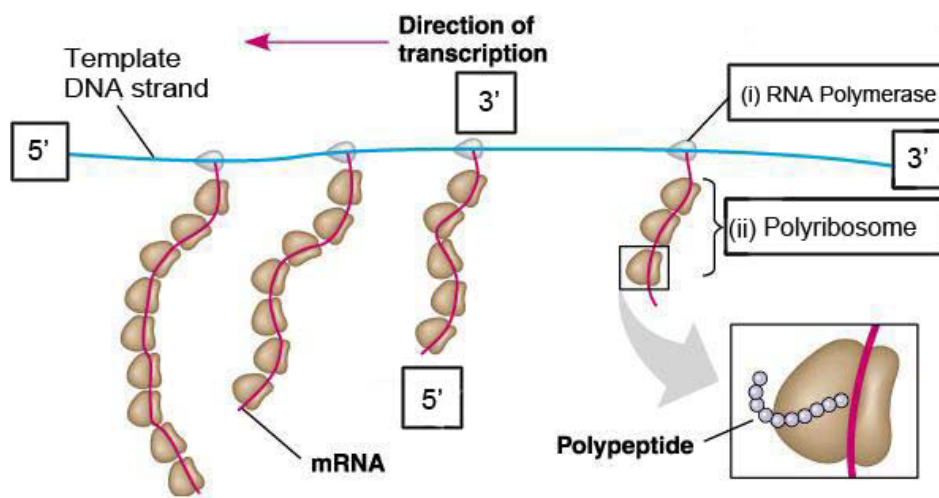


Fig. 1.2

- (d) Label the 5' and 3' ends of the mRNA and DNA template strand and structures (i) and (ii) on Figure 1.2. [4]
- (e) State the main difference between the processes shown in Fig. 1.2 and that occurring in eukaryotes. [1]
- Translation of one mRNA strand can occur simultaneously/concurrently with transcription of the same strand / coupled transcription-translation in prokaryotes but not in eukaryotes.
- (f) Explain your answers in (e). [3]
- Eukaryotic DNA contains introns which need to be removed before translation (but prokaryotes do not) / exons need to be spliced before translation.
[A: reverse argument]
 - Nucleus present / absent
 - [Elaboration]: Transcription in eukaryotes takes place in the nucleus while translation occurs in the cytoplasm / RER.

- mRNA has to travel from the nucleus to the ribosomes in the cytoplasm / RER for translation.

[Total: 14]

QUESTION 2

Phosphofructokinase (PFK) is a tetrameric enzyme that plays a central role in controlling the rate of glycolysis. In many organisms, the activity of PFK is enhanced allosterically by several substances, including ADP, and is inhibited allosterically by several other substances, including ATP and citrate. Fig. 2.1 shows part of the glycolysis reactions.

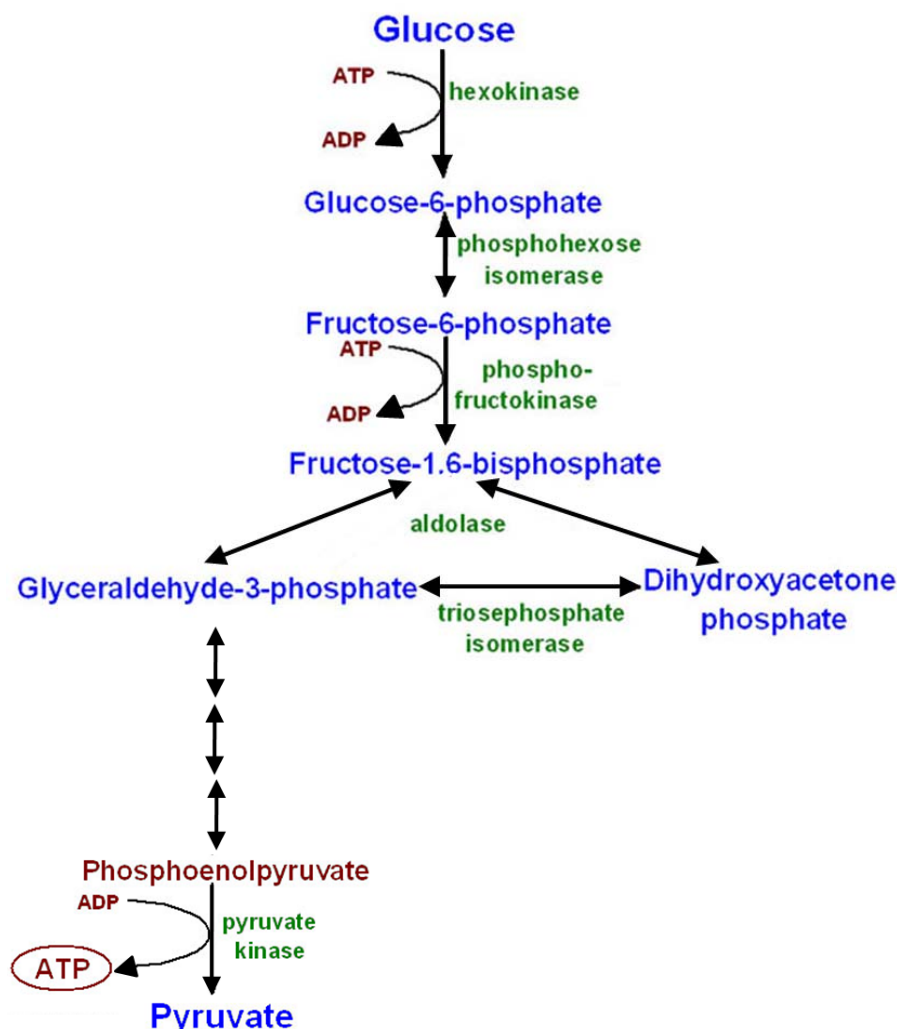


Fig. 2.1

(a) Explain why phosphofructokinase is a quaternary protein. [1]

- Consists of four polypeptide chains / subunits

(b) Besides PFK, hexokinase and pyruvate kinase also play a role in determining the overall rate of glycolysis.

With reference to Fig. 2.1, suggest why reactions catalyzed by these three enzymes determine the rate of glycolysis. [1]

- These reactions are irreversible.

(c) Explain the importance of inhibition of PFK by citrate when citrate concentration in the cell is high. [2]

- Product inhibition / feedback inhibition / allosteric inhibition
- Prevents cells from wasting chemical resources/energy / synthesizing more citrate/intermediate/ATP than needed
[A: reverse argument]
- Control of enzymatic action helps to maintain a stable internal environment in living organisms

(d) Describe how citrate inhibits PFK. [3]

- Citrate binds to the allosteric site of PFK.
[R: another site other than the active site]
- Induces a conformational change in PFK, affecting/changing/distorting the shape of the active site of PFK.
- Active site of PFK is no longer complementary to F-6-P and ATP
- [Stabilizes the inactive form of PFK]

(e) In mammals, under anaerobic condition, both Krebs cycle and oxidative phosphorylation do not occur. Instead, lactate fermentation occurs.

Explain the significance of lactate fermentation. [2]

- This step regenerates NAD...
- ...so that glycolysis can continue to produce ATP for cellular reactions to occur.
- ATP is produced via substrate level phosphorylation.

(f) Hexokinase, the enzyme involved in the first step of glycolysis, is able to utilize various hexoses such as glucose, fructose and mannose.

Suggest how hexokinase is able to utilize different substrates. [2]

- The binding of these substrates to the active site induces a conformational change so that the active site fits the substrate precisely.
- This can be explained by the Induced fit hypothesis...
[R: because of induced fit hypothesis, hexokinase can bind....] → poor phrasing!

Note: the context of question (f) is not just about the substrate being complementary to the active site, but rather, about how upon substrate binding, the shape of the active site changes to fit the substrate more snugly.

[Total: 11]

QUESTION 3

HIV is a retrovirus known to cause acquired immune deficiency syndrome (AIDS).

(a) Outline the process by which HIV enters the host cell. [3]

- Glycoproteins gp41 and gp120 on the HIV envelope bind to the CD4 cell surface receptors on the surface of T cells.
- Once bound, the viral envelope fuses with the plasma membrane of host cell...
- ...and releases the nucleocapsid into the host cell cytoplasm.

Long terminal repeats (LTRs) are sequences of DNA that are repeated hundreds or thousands of times. They are found in retroviral genetic material, flanking functional genes. In viruses like HIV, LTRs are necessary for the insertion of viral genetic sequences into host genomes using the enzyme integrase. HIV-1 integrase binds to specific sites at the ends of the viral DNA.

Secondly, LTRs act as promoters that can control transcription when integrated into the host genome. Protein-binding sites in the LTR are involved with transcription. Fig. 3.1 shows a basic model of a retroviral genome.

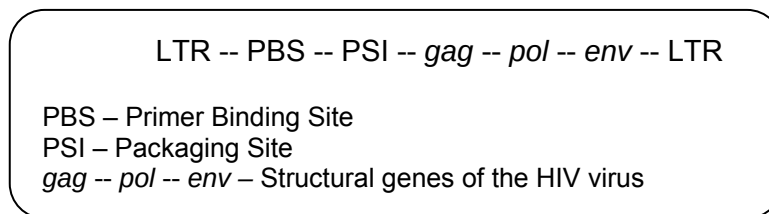
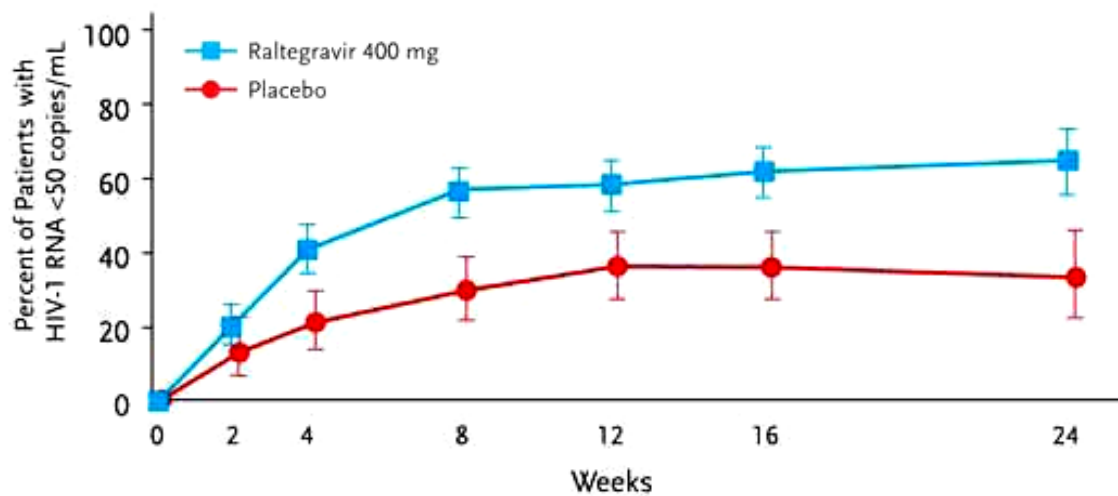


Fig. 3.1

(b) Explain how LTR sequences might function as a promoter for transcription. [2]

- The LTR sequences may serve as recognition sites for basal/general transcription factors.
- This is possible as the LTR sequences are complementary to the binding site of transcription factors.

Many integrase inhibitors have been discovered during the past 10 years, with some of them presently in clinical trials. An example of one such inhibitor is Raltegravir. In one such trial, one group of patients was given Raltegravir while the control group was given placebos (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are shown in Fig. 3.2.



Adapted from Merck & Co., Inc. Results of BENCHMRK-1 and-2, two phase-III studies evaluating the efficacy and safety of raltegravir, a novel HIV-1 integrase inhibitor, in patients with triple-class resistant virus. 14th Conference on Retroviruses and Opportunistic Infections; February 25–28, 2007; Los Angeles, CA.

Fig. 3.2

(c) With reference to Fig. 3.2, describe the effect of treating patients with 400mg of Raltegravir. [2]

- The groups of patients treated with Raltegravir had a higher percentage of patients showing HIV-1 RNA < 50 copies per ml as compared to the control group.
- The percentage of patients with HIV-1 RNA < 50 copies /ml increased from 20% in week 2 to 60% in week 24 in the group undergoing treatment whereas the percentage of patients with HIV-1 RNA < 50 copies /ml in the control group increased from 10% in week 2 to 30% in week 24.

(d) Suggest how Raltegravir might work as an antiretroviral drug. [2]

- Raltegravir inhibit the integration of HIV DNA into the human genome / prevents formation of provirus.
- HIV can no longer undergo replication / synthesise viral proteins / new viral particles / interfering with the life cycle of the virus and reducing infection.

Another result of HIV infection is cancer. Individuals infected with HIV have increased risk of three types of tumours: Kaposi's sarcoma, carcinomas of mouth and rectum, and B cell lymphomas disorders.

(e) Explain how a virus like HIV may cause cancer. [3]

- Insertion of viral DNA is random.
- Insertion at proto-oncogenes may result in the formation of oncogenes...
- ...which code for hyperactive protein / excessive proteins / a gain-of-function mutation / causes excessive cell division
- Insertion at the tumour suppressor genes / disruption of tumour suppressor genes...
- ...would produce non-functional / abnormal protein products / a loss of function mutation / the protein is unable to inhibit cell division.

[Total: 12]

Flower color in *Dahlia* flowers is determined by two gene loci, **A/a** and **B/b**.

A yellow **a** white
B white **b** yellow

Yellow-flowered plants	67
White-flowered plants	299

Parental phenotype: White-flowered plants x White-flowered plants
Parental genotype: AaBb x AaBb

	AB	Ab	aB	ab
AB	AABB White	AABb White	AbBB White	AbBb White
Ab	AABb White	AAbb Yellow	AaBb White	Aabb Yellow
aB	AaBB White	AaBb White	aaBB White	aaBb White
ab	AaBb White	Aabb Yellow	aaBb White	aabb White

Gametes: [1]
Genotype: [1]
Phenotype: [1]

F₁ phenotypic ratio: White flowers : Yellow flowers
 13 : 3

10

(b) Explain how interaction between the two genes gives rise to white-flowered plants. [4]

[Max 2]

- Dominant epistasis
[R: epistasis / gene interaction]
- The dominant allele B is epistatic to the A/a gene locus
[A: reverse argument: hypostatic]
- As long as one copy of allele B is present, regardless of the genotype at the locus A/a, the flower will be white

[Max 2]

- Allele A codes for a functional enzyme that synthesizes the yellow pigment.
- Allele a codes for a non-functional enzyme that is unable to synthesize the yellow pigment.
- Allele B codes for a protein that inhibits the expression of locus A/a / gene A.

(c) Explain the purpose of performing a chi-squared test on the results of the cross. No calculation is required. [2]

- To determine if the difference between the observed and expected results is significant / due to chance.
- If the probability value is more than 0.05, the difference is insignificant.
OR
If the probability value is less than 0.05, the difference is significant.

(d) Explain how you could determine that a yellow-flowered plant selected from the above cross is pure-breeding. [3]

- Test cross the yellow-flowered plant with a white-flowered plant which is homozygous recessive at both loci / aabb.
- If the yellow-flowered plant is not pure breeding (Aabb), the offspring ratio would be 1 yellow-flowered plant : 1 white-flowered plant.
- If the yellow plant is pure breeding (AAbb), the offspring will be all yellow-flowered plants.

[Total: 14]

QUESTION 5

Bacterial conjugation was first discovered in 1945 when Joshua Lederberg began investigating mating in bacteria. He conducted the following experiment as in Table 5.1 and Fig. 5.1.

Strain A, B and a mixture of both strains were grown, washed in minimal medium (MM) and grown on culture plates containing minimal medium that does not contain essential amino acids. The results are shown in Fig. 5.1.

Mutant genes (–) do not code for enzymes that synthesize amino acids. Note that all five amino acids are required for bacterial growth.

Bacterial strains	Genes for biosynthesis of amino acids	Mutant genes for biosynthesis of amino acids
A	thr ⁺ leu ⁺ thi ⁺	met [–] bio [–]
B	met ⁺ bio ⁺	thr [–] leu [–] thi [–]
Mixtures of Strains A and B	As found in strains A and B	As found in strains A and B

Table 5.1

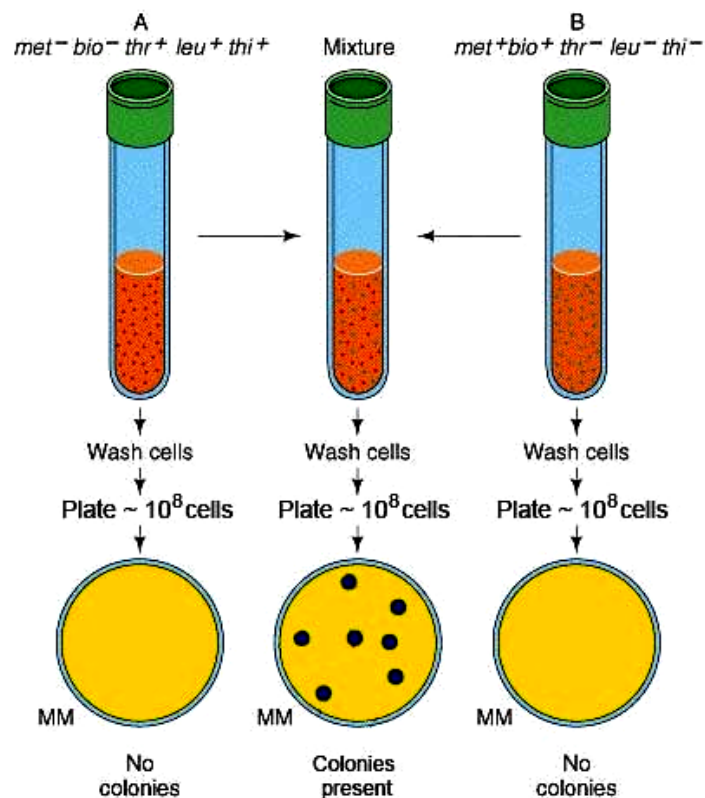


Fig. 5.1

(a) From the experiment in Fig. 5.1, explain the results obtained for

(i) strain A [2]

- There were no colonies present on MM plates as the bacteria lacked two essential amino acids / met and bio / all 5 amino acids
- This was because Strain A bacteria lacked the genes met⁺ and bio⁺ that code for the enzymes that synthesise the respective amino acids.

(ii) mixture of strain A and B [3]

- There were colonies present on MM plates as the bacteria could synthesize / produce / make essential amino acids.
- This was because Strain A and B bacteria underwent conjugation and genetic recombination.
- Resulting in a strain of bacteria with the genes met⁺, bio⁺, thr⁺, leu⁺ and thi⁺ that code for the enzymes that could synthesise all essential amino acids.

(b) Name one other factor that must be present in either Strain A or B for these results to be possible. [1]

- The F plasmid / fertility factor F.

(c) Lederberg and his collaborators later discovered viral transduction in 1951.

State three differences between transduction and conjugation. [3]

Feature	Transduction	Conjugation
Involvement of viruses	Transduction requires the <u>involvement of a lysogenic phage</u> .	Conjugation <u>does not involve viruses</u> .
Involvement of a sex pilus	Transduction <u>does not involve</u> transferring genetic material via a <u>sex pilus</u> .	Conjugation involves transferring genetic material via a <u>sex pilus</u> .
Transfer of DNA	DNA is <u>transferred by / via bacteriophages</u> from one bacterial cell to another	DNA is <u>transferred via a sex pilus</u> formed between two bacterial cells
DNA	Genes / DNA <u>from the bacterial genome</u> is transferred	<u>F plasmid</u> / Genes <u>from F plasmid</u> is transferred
DNA	In <u>specialised</u> transduction, both <u>viral genome and part of the bacterial genome</u> is transferred OR In <u>generalised</u> transduction, <u>part of the bacterial genome</u> is transferred.	In conjugation, <u>only F plasmid</u> is transferred <u>but not the bacterial genome</u>

<u>Transfer of DNA</u>	Generalised transduction - Transfer of a <u>random</u> piece of bacterial DNA	Transfer of <u>F plasmid</u> (more specific)
Insertion of genetic material	<u>Specialised transduction</u> involves the <u>insertion of genetic material</u> / genes / <u>bacterial and viral</u> DNA <u>into the host genome</u> to form a prophage.	The <u>F plasmid</u> is <u>not integrated</u> the host genome.
Lysis of donor cell	Transduction involves the <u>lysis</u> of the original bacterial cell/donor cell/cell that donates genetic material.	Conjugation <u>does not result in the lysis</u> of cells.

[Total: 9]

QUESTION 6

Fig. 6.1 shows the light-dependent stage of photosynthesis.

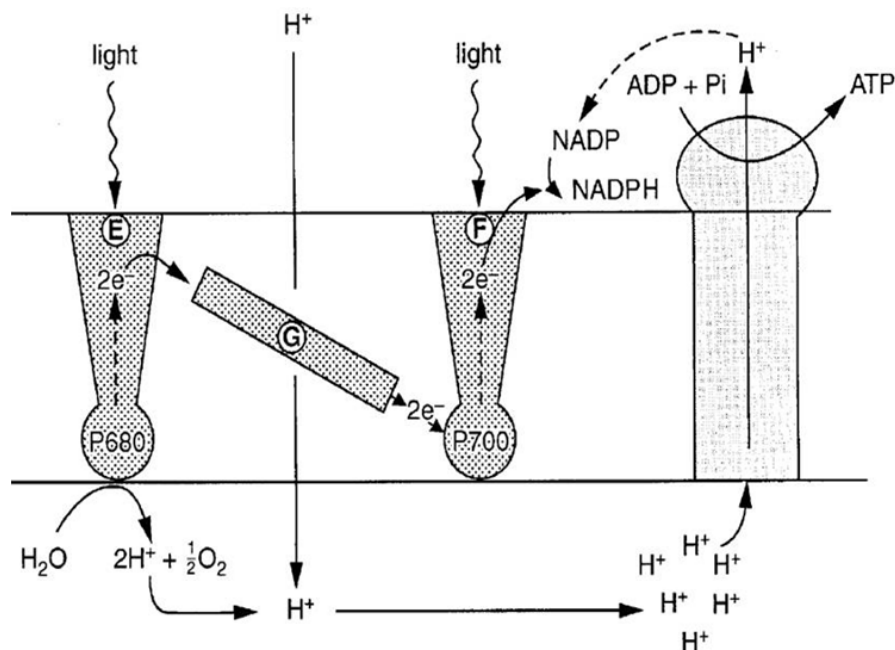


Fig. 6.1

(a) Diquat and paraquat are herbicides (weed killers) that readily accept electrons. They are known to act on the thylakoid membranes.

(i) With reference to Fig. 6.1, explain how these herbicides cause the plants to die. [5]

- No photophosphorylation
- Excited electrons emitted from P680 and P700 / special chlorophyll a / photosystem are accepted by the herbicide
- No electrons flow along the electron transport chain / electron transport chain halted
- No ATP and NADPH produced
- Calvin cycle ceases OR dark reaction ceases OR No production of organic molecules / sugars / triose / GALP / glucose
- Respiration cannot occur since there is no respiratory substrate, leading to cell death.

(ii) Some weed species are not killed when these herbicides are applied to them.

Suggest why this is so.

[1]

- Herbicide-resistant gene / are herbicide resistant
- Possess enzymes to break down / inhibit the herbicide
- Possess membrane pumps to actively pump out herbicides that enter the cell
- Have electron acceptors which have higher affinity than herbicide
- The cell membrane is not permeable to the herbicide.
- Cuticle is thick, herbicide cannot enter.

(iii) Describe one distinct structural difference between a phospholipid molecule found in the thylakoid membrane and a fat molecule. [1]

- Phospholipid comprises one phosphate group and two fatty acids whereas a fat molecule comprises three fatty acids.
OR
- Phospholipid comprises one phosphate group and two fatty acids attached to glycerol whereas a fat molecule comprises three fatty acids attached to glycerol.

(b) Plants that live in the Arctic have a relatively short growing season in which the light intensity is always relatively low. Many species growing in these conditions have a high level of anthocyanin, an accessory pigment, in their leaves. The combined effect of these red pigments with the green chlorophylls makes the leaves appear dark purple or black.

Suggest why this adaptation is useful in increasing photosynthetic rates. [3]

Light reaction

- Presence of anthocyanin increases the range of wavelengths / colors of light that the plant may absorb
- Maximize absorption as light intensity is always relatively low
- More energy available for photoactivation / excitation of electrons
- More ATP and NADPH formed for

Dark reaction

- Increases temperature of the leaves (dark color absorbs heat better)
- Enzymes in the Calvin cycle function more efficiently
- AVP

[Total: 10]

QUESTION 7

Mitochondrial DNA analysis of various rodents was done and their phylogenetic relationships represented by Fig. 7.1

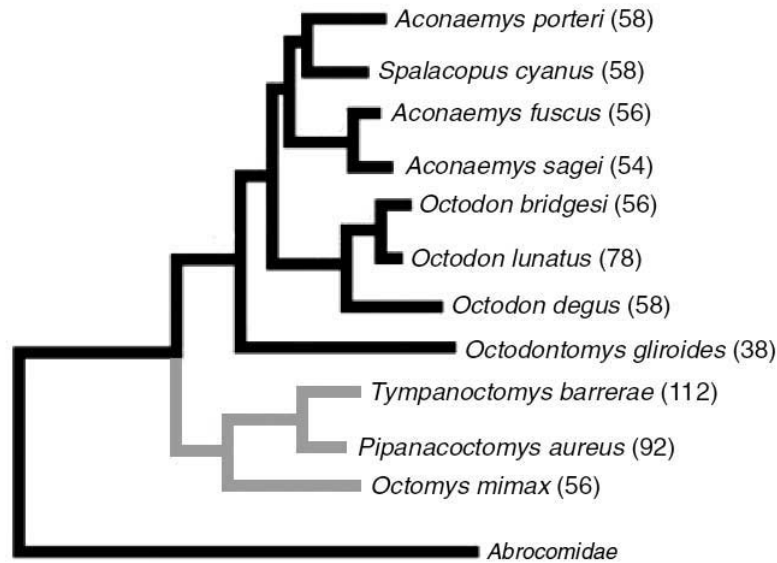


Fig. 7.1

(a) Suggest why mitochondrial DNA was used to study the phylogenetic relationships between the various species of rodents. [2]

- Mitochondrial DNA is a homologous molecule/nucleotide sequence...
- ...that is found in the mitochondria of every eukaryotic organism.

(b) Explain how DNA analysis of mitochondrial DNA can provide evidence for the relationships shown in Figure 7.1. [2]

- DNA-DNA hybridization can be used to examine the phylogenetic relationships between species.
- The more complementary the nucleotide sequences, the more closely related they are. [A: reverse argument]

OR

- DNA analysis / compare sequence of mitochondrial DNA of all species of rodents.
- The greater the differences in nucleotide sequence, the more distantly related the species are. [A: reverse argument]

Cases of mammalian tetraploidy are rare. The Plains Viscacha Rat, *Tympanoctomys barrerae*, is one of the few polyploid mammals in existence and it is tetraploid (4n) with 112 chromosomes. It is only found in the desert scrublands of the central western plains in Argentina. It was hypothesised that *Tympanoctomys barrerae* evolved from the same common ancestor as *Octomys mimax*, a diploid rodent with 56 chromosomes (2n). *O. mimax* inhabits rocky desert and semi-desert environments up to 800 metres above sea level.

(c) Explain how tetraploidy may have first developed in the ancestral rat population. [2]

- Spindle fibres may not have formed during prophase I of meiosis.
- Cytokinesis may not have occurred during meiosis.
- The fusion of two diploid gametes could have given rise to the tetraploid rat.
- Causing the doubling of chromosomes from 56 (2n) to 112.

(d) Describe how *Tympanoctomys barrerae* may have evolved as a separate species as a result of natural selection. [4]

- Genetic variation/tetraploidy existed in the original population as a result of mutation.
- In certain environments like the plains / desert scrubland / lowland / non-mountainous desert regions, it was selectively advantageous to be tetraploid and the rats survived to reproductive age and produced fertile and viable offspring.
- The rats passed on the tetraploid chromosome number to their offspring / the offspring also are tetraploids / have a tetraploid chromosome number.
- Over time, the tetraploid rats predominated in the population.
- Accumulation of other advantageous alleles occurred over time.
- The rat population underwent speciation / became a separate species as they could no longer interbreed with the original population.

[Total: 10]

QUESTION 8

(a) Describe the molecular structure of cellulose found in the cell wall of a plant. [6]

1. Made up of β -glucose monomers.
2. One glucose molecule is rotated 180° and is upside down relative to the adjacent molecule...
3. ...to give a β -1,4 glycosidic bond.
4. Straight and unbranched.
5. The hydroxyl groups of one chain form hydrogen bonds with hydroxyl groups of neighbouring chains...
6. ...resulting in the formation of cross-links.
7. In the cell wall, groups of about 60 to 70 cellulose molecules become tightly cross-linked to form bundles called microfibrils.
8. Microfibrils are in turn held together in bundles called fibers by hydrogen bonding.

(b) Explain how ATP enables nerve cells to transmit nerve impulses. [6]

1. Used by the sodium-potassium pump
2. Pump 2 potassium ions in and 3 sodium ions out, both against their concentration gradients
3. Maintain resting potential / restore resting potential after an action potential
4. Synthesis of acetylcholine from choline and ethanoic acid
5. Synthesis of acetylcholine-gated ion channels / acetylcholine receptors
6. Reabsorption of neurotransmitters into the synaptic knob
7. Movement of neurotransmitter-containing vesicles towards the cell membrane
8. Fusion of vesicles with the cell membrane
9. Used by the calcium pump to pump calcium ions out of synaptic knob / neurone to restore low cytosolic calcium ion concentration.

(c) Explain the homeostatic mechanism in response to a rise in blood glucose concentration. [8]

1. Homeostasis is the maintenance of constant / stable internal environment...
2. ...despite changes in external environment.
3. When blood glucose is raised beyond set point / 90mg glucose per 100ml blood...
4. ...negative feedback mechanism is activated.

5. Beta cells in the Islets of Langerhans / pancreas are stimulated to secrete insulin into the bloodstream.
6. Alpha cells stop secreting glucagon.
7. Target cells of insulin include liver and muscle cells
8. **Response:** Increased permeability of cell membrane to glucose
9. **Response:** Increased rate of oxidation / metabolism / utilization / use of glucose
10. **Response:** Increased glycogenesis / conversion of glucose to glycogen
11. Fall in blood glucose concentration back to set point...
12. ...which serves as a negative feedback to β -cells to stop secreting insulin.

QUESTION 9

(a) Describe the main structural features of the *lac* operon.

[8]

1. The *Lac* operon consists of a length of DNA sequence.
2. It also consists of a promoter,
3. which is a binding site for RNA polymerase.
4. It consists of a CAP/CRP binding site next to the promoter.
5. It consists of an operator,
6. which is a binding site for the repressor / inhibitor protein.
7. The operon consists of three structural genes, namely
8. Lac Z which codes for β -galactosidase,
9. Lac A which codes for lactose transacetylase,
10. Lac Y which codes for lactose permease.
11. It also consists of a termination sequence at the end of the last structural gene.

(b) Explain how mitosis maintains genetic stability.

[6]

1. Mitosis maintains genetic stability by ensuring that the two daughter cells contain exactly the same number of chromosomes as the parent cell.
2. Genetically identical

[Point 3-6 explains the features that ensure the chromosome constitution is maintained from one generation to the next]

3. The chromatin of the parental nucleus replicate during S phase of interphase before mitosis begins.
4. All chromosomes align at the equator during metaphase...
5. ...to ensure that the sister chromatids are shared equally between the two daughter nuclei after anaphase occurs.
6. The division of the centromere ensures that one sister chromatid of each chromosome will move to opposite poles.
7. There is no association between homologous pairs of chromosomes / bivalents do not form, therefore no chiasmata are formed at prophase I.
8. There is thus no crossing over which would have resulted in genetic variation/recombination of alleles.

(c) Distinguish between the processes of meiosis I and meiosis II in eukaryotes.

[6]

	Meiosis I	Meiosis II
Prophase I VS Prophase II	• <u>Pairs of homologous chromosomes</u> are <u>present</u>. • <u>Homologous chromosomes pair up / synapsis</u> occurs to form <u>bivalents</u>. • <u>Chiasmata</u> form • <u>Crossing over</u> takes place	• <u>One from each pair of homologous chromosome</u> is <u>present</u>. [R: homologous chromosome absent] • Hence, <u>no pairing up / synapsis / formation of bivalents</u> occurs. • <u>No chiasma</u> form • <u>No crossing over</u>
Metaphase I VS Metaphase II	• <u>Homologous chromosomes line up in two rows</u> along the <u>equator / metaphase plate</u>.	• <u>Chromosomes line up in a single row</u> along the <u>equator / metaphase plate</u>.
Anaphase I VS Anaphase II	• <u>Homologous chromosomes separate</u>. • <u>No division of centromeres</u>.	• <u>Chromatids separate</u>. • <u>Centromeres divide</u>.
Ploidy	• Cells that are <u>initially diploid</u> are <u>haploid</u> by the end of meiosis I.	• Cells start out <u>haploid</u> and <u>remain haploid</u> at the end of meiosis II.

[Total: 20]

● End of Paper 2 ●